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Capstone Project- Business Analytics Final Project Proposal

Analyzing Factors Influencing Digital Banking Adoption

I chose this business problem because I have approximately five years of experience working in the banking industry, where I was directly involved in the transition from a traditional banking system to a more modern digital banking platform. During this transition, I served as the digital banking ambassador for my branch, which gave me firsthand experience assisting customers with adapting to new digital services such as online banking, mobile banking, and digital account management tools.

Through this experience, I observed that customer reactions to the transition varied significantly across different demographic groups. While many customers appreciated the convenience and efficiency of the new platform, others, particularly older customers, expressed hesitation, frustration, or difficulty adapting to the technology. Some customers were resistant to change due to concerns about security, lack of familiarity with digital tools, or preference for in-person banking interactions.

This experience sparked my interest in understanding the factors that influence digital banking adoption and customer satisfaction. I believe this topic is both relevant and valuable because financial institutions continue to invest heavily in digital transformation initiatives, yet customer adoption remains a major challenge. By analyzing customer behavior and the factors affecting

mobile or digital banking adoption, this project could provide insights that help banks improve user experience, increase adoption rates, and better support customers during technological transitions.

Potential obstacles may include limited dataset size, missing values, or limitations in demographic diversity within the dataset. Additionally, customer behavior and digital banking adoption may be influenced by external factors not captured in the dataset, which could affect predictive accuracy.

Dataset Source

The dataset I plan to use is the “Factors Affecting Mobile Banking Adoption” dataset from [Kaggle](#)

Potential variables

Potential variables I may analyze include:

- Age
- Trust and security concerns
- Ease of use
- Income
- Education level
- Digital literacy
- Mobile usage habits

Analytical Methods

The analytical methods I may include are:

1. **Classification Models**

Possible models include logistic regression and decision trees to predict the likelihood of digital banking adoption.

2. **Clustering**

Customer segmentation techniques may be used to identify groups such as highly digital users, hesitant users, low-tech users, and security-concerned users.

3. **Regression Analysis**

Regression models may be used to predict digital engagement levels, adoption likelihood scores, or usage frequency.

4. **Dashboard Visualization**

Tableau dashboards may be created to visualize adoption trends, demographic insights, influencing factors, and customer segments.

Project Structure

I will be completing this project individually.